

Alain Emmanuel P. Cheng

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EDUCATION

De La Salle University

Bachelor's of Science in Computer Science major in Software Technology

NCR, Manila

Graduated April 2026

SKILLS

Languages: Javascript, C#, Python, Typescript, C/C++, Java, Rust

Frameworks & Libraries: React, Tailwind, ASP.NET Core, Next.js, Material UI, Node.js, TensorFlow, PyTorch, Three.js, Framer Motion

Databases: Microsoft SQL Server, PostgreSQL, MySQL, MongoDB, SQLite

Tools: Git, SSMS, DbSchema, Azure DevOps, Docker, VS Code, Android Studio, Figma, Postman

EXPERIENCE

Software Engineer Intern

Elinnov Technologies, Inc.

September 2025 - December 2025

- Developed RESTful API endpoints using ASP.NET Core and implemented frontend user interfaces with React.
- Participated in an Agile software development team using Azure DevOps for project tracking and version control.

Volunteer Research Laboratory Representative

The Center for Computational Imaging and Visual Innovations

May 2024

- Represented our laboratory's booth by engaging with visitors in explaining key research projects and promoting the members' innovations and works.

PROJECTS

Hardware Inventory Management System

Personal Project

January 2026 - Present

- Developing a lightweight local desktop application intended for small business operations built with React and Tauri frameworks, Rust backend, and a MS SQL Server database.
- Designed modular frontend components that supports inventory, sales, purchases, and reporting workflows.

"Towards Improving StegaStamp's Detection of Hidden Messages in Images" — Machine Learning in Computer Vision

Undergraduate Thesis

September 2023 - August 2025

- Our team trained, evaluated, and benchmarked different semantic segmentation models using Tensorflow (Python) on NVIDIA Tesla V100 GPUs to improve detection of visually embedded messages in images.
- Designed MobileBiSeNet, an optimized variant of BiSeNet that achieved an average of 25.2 FPS, improving inference speed by 41% over the baseline while maintaining comparable segmentation accuracy.

G-Cakes — Web Application

Software Engineering Coursework

February 2023 - April 2023

- Developed an improved web app for a small local cake business using TypeScript and Next.js for the frontend while integrating Contentful CMS for content management and using Node.js for the backend. The application was deployed via Vercel.
- Our team managed the project using Jira and followed an Agile SDLC.

AWARDS AND CERTIFICATIONS

Academic Awards: First Honor Dean's Lister (AY 2023-2024, Term 3)

Certifications: Foundational C# with Microsoft (September 17, 2025)